CHAPTER 9

Dates and Times

Let's get one thing out of the way before we go any further: time is an absolute mess. Some physicists aren't even sure it's real; Einstein theorized – and it was later proved – that time (and space) distort with gravity and velocity; Douglas Adams proposed it was actually entirely bent. For crying out loud, we don't even measure it in Base 10! (Thanks, ancient Babylon.) Except that actually we do sometimes measure it in Base 10 because of science and computing. To give an extreme example of the difficulty of measuring time accurately in computer systems: the GPS (Global Positioning System) actually corrects for relativistic effects – a GPS satellite's internal clocks are tuned to be slightly slower on Earth so that when they're traveling at speed (Special Relativity), and in orbit (slightly weaker gravitational field, so General Relativity), they run at exactly the same speed relative to their counterparts on the ground. Without these corrections the GPS system would be less accurate.

Humans have been measuring time for thousands of years, mostly by observing the Sun. When the sun is at the highest point in the sky, we call it midday, but of course this is highly localized. Solar midday in London happens later than in, say, Berlin. Even within a single country, there can be a variance of several minutes between relatively nearby cities. This played havoc with early train timetables in the United Kingdom, but in 1884, we standardized on GMT (Greenwich Mean Time): we picked a point on the map and used that to set the time for the entire country. Indeed, GMT was an international standard until 1972. This idea of having a standard by which to reckon time across the world lives on in UTC (Coordinated Universal Time) and is the basis for how all computer systems measure and record time today. I'm sure anyone reading this book has at least heard of time zones, which are simply offsets from UTC time that are used in different regions of the world: CET (Central European Time) is UTC + 1 hour, PST (Pacific Standard Time) is UTC – 8 hours, and so on. We'll be having some fun with time zones in due course. Plus or minus an hour or so…

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